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Internship Track:	Cyber Security Intern
Submission Date:	22 October
Week :	1
Task 1:	Lab Setup,Linux and Network Foundation
Student Id:	DG/15/OCT/CYBER/118

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Virtual Machine Networking Modes

1. Introduction

Virtual machines use virtual network adapters to communicate with their host, other virtual machines or external networks. The selected networking mode determines which devices can communicate and what network exposure the VM has.

2. Comparison

Mode	How it works	Appropriate use	Main consideration
NAT	A VM typically accesses external networks through address translation performed by the virtualization platform.	Updates and internet access for an isolated lab machine.	Other machines may not directly initiate connections to the VM without additional configuration.
NAT Network	VMs on the same configured NAT network can communicate with each other and access external networks through NAT.	A small lab requiring VM-to-VM communication and internet access.	Verify network isolation and the platform's routing/firewall configuration.
Host-only	VMs communicate on a private virtual network with the host; internet access is generally absent unless explicitly configured.	Isolated labs for controlled practice.	Host-only does not guarantee isolation from the host or other host-connected services.
Bridged	A VM connects to the physical LAN as though it were another device on that network.	Authorized testing where a VM must participate in the real LAN.	Can expose vulnerable VMs to real devices and networks.

3. Why Vulnerable Labs Must Not Be Bridged

Intentionally vulnerable machines may contain weak credentials, exploitable services or insecure configurations. Bridging such a machine can expose it to other devices on the physical network and potentially the internet.

For beginner labs:

- Prefer Host-only networking when internet access is unnecessary.
- Use a dedicated NAT Network only when inter-VM communication and controlled internet access are needed.
- Do not enable port forwarding unless required and understood.
- Keep vulnerable targets separate from production systems and personal devices.
- Verify the actual connectivity and firewall rules before running exercises.

4. Suggested Lab Design

Kali Linux is the testing workstation, Ubuntu is the Linux practice machine, and Windows evaluation is the Windows practice machine. Connect them to an appropriately isolated virtual network for authorized lab exercises.

Record the selected mode, subnet, gateway, DNS configuration and actual IP address of each machine.

5. Snapshots

Take a clean snapshot after installing and updating each VM. Snapshots allow a VM to be restored to a previous state, but they are not a substitute for separate backups.

6. Conclusion

Virtual networking modes are an important part of lab safety. Choosing the correct mode reduces accidental exposure and makes controlled testing easier.